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(57) Abstract:

A shape memory alloy-based device with enhanced rotational motion (100) for converting reciprocating motion into rotational motion that includes a housing (102) with a base platform (104) and is a rotary member pivotally coupled on the base platform (104) and configured to move between a first and a second position; at least one connecting link (108) coupled to the rotary member (106) and a coil holder (110) through an attachment means (112); a thermally sensitive coil (114) positioned between the coil holder (110) and the rotary member (106), contracting and expanding with temperature changes induced by cooling (116) and heating units. The coil holder (110) is configured on guide bars (118), enabling slidable movement between the two positions which is transferred to the rotary member (106) through the connecting link (108), converting the linear motion of the coils (114) into rotational motion with increased torque and increased frequency.

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

&

The Patent Rules, 2003

COMPLETE SPECIFICATION

(See sections 10 & rule 13)

1. TITLE OF THE INVENTION

A SHAPE MEMORY ALLOY-BASED DEVICE WITH ENHANCED ROTATIONAL MOTION

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure relates to the field of mechanical systems and more particularly, the present disclosure pertains to a device with enhanced rotational motion using a shape memory alloy coil (SMA).

BACKGROUND

[0002] In recent years, there has been increasing interest in utilizing thermal energy for mechanical applications. Thermally responsive materials, such as shape memory alloys or thermally sensitive coils, offer promising alternatives for converting thermal energy into mechanical motion. These materials exhibit thermo-mechanical properties that are useful in constructing thermally actuatable devices. Generally, a shape memory alloy (SMA) is a metallic alloy that has distinctly different phases on opposing sides of a transition temperature. An SMA reaches a first physical state when it is below its transition temperature and a second physical state when it is above its transition temperature. The warmer second physical state is generally thought to have higher degrees of solid-phase crystalline order and symmetry than the cooler first physical state. The first and second states are typically referred to respectively as martensitic and austenitic states or phases in literature related to SMA technologies.

[0003] The most commonly used SMA appears to be an alloy called Nitinol that is approximately fifty-six percent (by weight) nickel and forty-four percent titanium. Other available shape memory alloys include copper-zinc-aluminum and copper-aluminum-nickel. The transition temperature of an SMA is highly sensitive to the composition of the alloy and can be selected by slightly varying the constituent ratios. Transition temperatures for Nitinol can reportedly be selected between negative one hundred and fifty degrees Celsius and positive one hundred and fifty degrees Celsius. Transformations typically take place over several minutes of time.

[0004] However, existing systems that utilize thermally responsive materials for motion conversion are often limited in their efficiency and effectiveness. Many such systems struggle with issues related to the precise

control of motion, the speed of response to temperature changes, and the durability of the components involved.

[0005] There is, therefore, a need for an improved device that uses shape memory alloy coil (SMA) for converting reciprocating motion into rotational motion by changing temperature of the SMA coil through cooling and heating units.

OBJECTS OF THE PRESENT DISCLOSURE

[0006] A general object of the present disclosure is to provide an improved device that converts reciprocating motion into rotational motion in higher torque.

[0007] Another general object of the present disclosure produce higher actuation frequency by reducing actuation timing of the SMA or thermal sensitive coil.

[0008] Yet another object of the present disclosure is adapted for various industrial applications, including power generation, robotics, and automated machinery.

[0009] Yet another object of the present disclosure is to provide improved device that has high power to weight ratio.

[0010] Yet another object of the present disclosure is to minimize noise for being adapted to sensitive environments.

SUMMARY

[0011] Aspects of the present disclosure relate mechanical systems and, more particularly, the present disclosure pertains to a device with enhanced rotational motion using a shape memory alloy coil (SMA).

[0012] According to an aspect, a device with enhanced rotational motion using a shape memory alloy coil (SMA) includes a housing having a base platform and a rotary member pivotally coupled to the centre of the base platform, the rotary member having at least one free end. The rotary member is configured to move between a first position and a second position. Additionally, the device

includes at least one connecting link having a first end and second end. The connecting link is coupled to at least one coil holder through an attachment means. Further, the device includes a plurality of thermally sensitive coils. The plurality of thermally sensitive coils are coupled between the at least one coil holder and the rotary member. The plurality of thermally sensitive coils are configured to expand in response to increase in temperature of the thermally sensitive coils by a heating unit. The device includes a cooling unit. The cooling unit is positioned on the base platform. The cooling unit is configured to spray a cooling fluid onto the heated thermally sensitive coils. The cooling fluid is spayed through the one or more humidifier units. The cooling unit contracts the expanded thermally sensitive coils with increased frequency. The cooling unit reduces the cooling time of the thermally sensitive coil. The cooling unit improves the actuation timing of the device.

[0013] The at least one coil holder is configured on one or more guide bars. The one or more guide bars allows slidable movement between the first position and the second position in response to the contraction and expansion of the plurality of thermally sensitive coils. The movement of the at least one coil holder being transmitted to the rotary member through the at least one connecting link. The plurality of thermally sensitive coils and the cooling unit facilates the rotation of the rotary member with increased torque and increased frequency.

[0014] In an embodiment, the rotary member may include at least two free ends and the at least one connecting link comprises at least a pair of connecting links, wherein each of the free ends of the rotary member is coupled to the first end of each of the at least one connecting link.

[0015] In an embodiment, the at least one coil holder may include a first coil holder and a second coil holder, each coil holder rotatably coupled to the second end of each of the at least one connecting link for converting reciprocating movement of the at least one coil holder into rotary motion using the at least one connecting link.

[0016] In an embodiment, the device may include a first set of bias springs positioned parallel to the plurality of thermally sensitive coils. The first set of bias

springs may include a first end and a second end. The first end of the first set of bias springs may be coupled to the first coil holder and the second end of the first set of bias springs may be coupled to the second coil holder through one or more strings. The first set of bias springs is configured to restrain the first and second coil holders in the second position due to change in the state of thermal-sensitive coil.

[0017] In an embodiment, the device may include one or more bias springs configured on the one or more guide rods. The bias springs include a first end and a second end. The first end of the bias springs may be coupled to the first coil holder and the second end of the bias springs may be coupled to the rotary member. The one or more bias springs may be configured to restrain the first and second coil holder in the second position due to temperature changes in the thermally sensitive coil. The string may be supported by one or more pulleys positioned on the base platform. The one or more pulleys may be configured to accommodate a portion of the string to facilitate the movement of the coil holder.

[0018] In an embodiment, the heating unit may be located in the housing and configured to supply power to the thermally sensitive coil to contract and facilitate movement of the first and second coil holders towards the first position of the coil holder and the rotary member.

[0019] In an embodiment, the at least one connecting link may include a first end and a second end. The first end of the at least one connecting link may be coupled to a free end of the rotary member. The second end of the at least one connecting link may be coupled to at least one coil holder through an attachment means.

[0020] In an embodiment, the cooling unit may include a first cooling unit and a second cooling unit. The a first cooling unit and a second cooling unit.may be fluidically coupled to a cooling fluid tank through one or more conduits.

[0021] In an embodiment, the rotary member may include an output shaft configured to rotate along with the rotary member. The output shaft may be coupled to a load or device for transferring rotational motion.

[0022] Various objects, features, aspects, and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The accompanying drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0024] FIG. 1A illustrates an exemplary isometric view of the device with an integrated lower cooling unit, according to the embodiments of the present disclosure.

[0025] FIG. 1B illustrates an exemplary isometric view of extension spring-based assembly in the device, according to the embodiments of the present disclosure.

[0026] FIG. 1C illustrates an exemplary isometric view of the SMA rotary actuator with an integrated Upper evaporative (spray) cooling system, according to embodiments of the present disclosure.

[0027] FIG. 1D illustrates an exemplary isometric view of an extension spring-based chassis device in an un-actuated state, according to embodiments of the present disclosure.

[0028] FIG. 1E illustrates an exemplary isometric view of the extension spring-based chassis device in an actuated state, according to embodiments of the present disclosure.

[0029] FIG. 1F illustrates an exemplary side view of the proposed device in an un-actuated state, according to embodiments of the present disclosure.

[0030] FIG. 1G illustrates an exemplary diagram of the proposed device showing compression spring-based device incorporates sma coils to structure

similar ton human muscle fibers to generate forces in an un-actuated state, according to embodiments of the present disclosure.

[0031] FIG. 1H illustrates an exemplary diagram of the proposed device showing the rotary member mechanism, according to embodiments of the present disclosure.

[0032] FIG. 1I illustrates an exemplary diagram of the proposed device showing a cam follower mechanism, according to embodiments of the present disclosure.

[0033] FIG. 1J illustrates an exemplary isometric view of the proposed device with a lower side cooling unit without a top cover and base platform, according to embodiments of the present disclosure.

[0034] FIG. 1K illustrates an exemplary top view of the lower side cooling unit integrated without top covering of the proposed device, according to embodiments of the present disclosure.

[0035] FIG. 1L illustrates an exemplary cross-section view of the proposed device with an integerated lower cooling unit, according to the embodiments of the present disclosure.

[0036] FIG. 1M illustrates an exemplary cross-section view of the proposed device with an integrated upper cooling unit, according to the embodiments of the present disclosure.

[0037] FIG. 2 illustrates an exemplary circuit diagram of the proposed device, according to embodiments of the present disclosure.

[0038] FIG. 3 illustrates an exemplary graph of the frequency of the SMA-based device under the effect of acetone cooling as compared to the case of free convection, according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0039] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such details as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the

contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the scope of the present disclosures as defined by the appended claims.

[0040] As used herein, “substantially” means largely or considerably, but not necessarily wholly, or sufficiently to work for the intended purpose. The term “substantially” thus allows for minor, insignificant variations from an absolute or perfect state, dimension, measurement, result, or the like as would be expected by a person of ordinary skill in the art, but that does not appreciably affect overall performance.

[0041] Throughout the present disclosure, “attachment means” includes, but is not limited to, screws, nails, rivets, adhesives, magnets, hook and loop fasteners, hook and slot fasteners, interlocking elements, friction-grip releasable fasteners, fastening straps, and the like.

[0042] Embodiments explained herein relate to the field of mechanical systems and, more particularly, the present disclosure pertains to a device for converting reciprocating motion into rotational motion using a shape memory alloy coil (SMA).

[0043] In an embodiment, a device incorporates SMA coils to structure similar tone human muscle fibers to generate forces, allow for the generation of actuation and output forces in the movable sub-assembly along the longitudinal axis. The device is connected to a modular transmission mechanism, such as a linkage and cam follower, which efficiently converts the longitudinal force from the coils into medium to high torque output. Simultaneously, strings are routed over pulleys, causing the bias springs to extend. This action stores energy, which is then used during the SMA contraction phase as they cool. The synergy between the SMA coils and the bias springs mimics an agonist-antagonist setup, thereby enhancing the actuation process such that the SMA coils and a cooling unit facilates the rotation of the cam with increased torque and increased frequency.

[0044] Referring to FIGs. 1A to 1M, where an exemplary isometric view of the proposed device for converting reciprocating motion into rotational motion is disclosed. The device 100 includes a housing 102 that includes a base platform

104 and a rotary member 106 pivotally coupled to the center of the base platform 104, the rotary member 106 having at least one free end 106-1. The rotary member 106 is configured to move between a first position and a second position. Additionally, the device 100 includes at least one connecting link 108 having a first end 108-1 and second end 108-2, wherein the first end 108-1 of the at least one connecting link 108 coupled to the at least a free end 106-1 of the rotary member 106 and the second end 108-2 of the at least one connecting link 108 coupled to at least one coil holder 110 through an attachment means 112. Further, the device includes plurality of thermally sensitive coils 114 coupled between the at least one coil holder 110 and the rotary member 106. The plurality of thermally sensitive coils 114 are configured to contract and expand in response to temperature changes induced by a cooling 116 and a heating unit. The plurality of thermally sensitive coils 114 are configured to be arranged in an array forming a muscle fibre structure. The at least one coil holder 110 is configured on one or more guide bars 118 that allow slidable movement between the first position and the second position in response to the contraction and expansion of the plurality of thermally sensitive coils 114. The movement of the at least one coil holder 110 is transmitted to the rotary member 106 through the at least one connecting link 108, thereby converting the linear motion of the thermally sensitive coils 114 into the rotational motion of the rotary member 106. In an embodiment, the device 100 may feature two versions an extensive spring mechanism and a compressive spring mechanism.

[0045] In an embodiment, the rotary member 106 may include at least two free ends 106-1, 106-2, and the at least one connecting link 108 comprises at least a pair of connecting links 108-1, 108-2. Each of the free ends 106-1, 106-2 of the rotary member 106 may be coupled to the first end 108-1A, 108-2A of each of the at least one connecting link 108 through the attachment means.

[0046] In an embodiment, the rotary member 106 may include a base member fixed at the center of the base platform 104. The base member may be coupled to the plurality of thermally sensitive coils 114. Further, the rotary

member 106 may include a cam for converting reciprocating motion into rotation motion using one or more connecting link 108.

[0047] In an embodiment, the at least one coil holder 110 may include a first coil holder 110-1 and a second coil holder 110-2, each coil holder 110 rotatably coupled to the second end 108-1B, 108-2B of each of the at least one connecting link 108 for converting reciprocating movement of the at least one coil holder 110 into rotary motion using the at least one connecting link 108.

[0048] In an embodiment, the device 100 may include a first set of bias springs 120 positioned parallel to the plurality of thermally sensitive coils 114, the first set of bias springs 120 having a first end 120-1 and a second end 120-2, the first end 120-1 of the first set of bias springs 120 coupled to the first coil holder 110-1 and the second end 120-2 of the first set of bias springs 120 coupled to the second coil holder 110-2 through one or more strings 122. The first set of bias springs 120 may be configured to restrain the first and second coil holders 110-1 and 110-2 in the second position due to temperature changes in the thermal sensitive coil 114.

[0049] In an embodiment, the one or more string 122 may be supported by one or more pulleys 124 positioned on the base platform 104. The one or more pulleys 124 may be configured to accommodate a portion of the string 122 to facilitate movement of the coil holder 110.

[0050] In an embodiment, the device 100 may include one or more bias springs 126 may be configured on the one or more guide rods 118. The bias springs 126 having a first end 126-1 and a second end 126-2. The first end 126-1 of the bias springs 126 may be coupled to the first coil holder 110-1 and the second end 126-2 of the bias springs 126 may be coupled to the rotary member 106. The one or more bias springs 126 may be configured to restrain the first and second coil holder 110-1, 110-2 in the second position due to temperature changes in the thermally sensitive coil 114.

[0051] In an embodiment, the heating unit may be located in the housing 102 and configured to supply power to the thermally sensitive coil 114 to contract

and facilitate movement of the first and second coil holders 110-1, 110-2 towards the first position.

[0052] In an embodiment, the cooling unit 116 may include a first cooling unit 116-1 and a second cooling unit 116-2. The first cooling unit 116-1 may be positioned on the sides of the base platform 104 and includes one or more humidifier units 128 fluidically coupled to a cooling fluid tank 130 through one or more conduits 132. The cooling unit 116 may be configured to spray cooling fluid onto the thermally sensitive coil 114 through the one or more humidifier units 128 to reduce the cooling time of the thermally sensitive coil 114, thereby improving the actuation timing of the device 100.

[0053] In an embodiment, the second cooling unit 116-2 may be configured within the housing 102 at the top and includes one or more humidifier units 134 fluidically coupled to a cooling fluid tank 136. The second cooling unit 116-2 may be configured to spray cooling fluid onto the thermally sensitive coil 114 through the one or more humidifier units 134 to reduce the cooling time of the thermally sensitive coil 114, thereby improving the actuation timing of the device 100.

[0054] In an embodiment, the rotary member 106 may include an output shaft 140 configured to rotate along with the rotary member 106. The output shaft 140 may be coupled to a load or device for transferring rotational motion.

[0055] In an embodiment, the device 100 may include four a top container 162 with cover 160 for coolant storage and dispersion through the humidifier module (128,134), the bottom cover 164 embedding the base platform 104, and at least two electronic compartments for sensors/controllers 170 and battery supply unit 180.

[0056] In an embodiment, the device 100 may include a control unit operatively coupled to the sensors 170, the battery supply unit 180, the cooling unit 116, and the heating unit. The control unit may be configured to actuate the heating unit to supply power to the thermally sensitive coil 114 which contracts due to temperature changes, in contraction the thermally sensitive coil 114 moves towards the rotary member 106. The contraction and expansion of the thermally

sensitive coil 114 facilitate the movement in the rotary member 106 to actuate the load or the device.

[0057] In an exemplary embodiment, the each thermally sensitive coil 114 of the plurality of thermally sensitive coil 114 may be a shape memory alloys (SMAs) coil a distinct category of active materials capable of recovering their original shape when heated to an elevated temperature phase, known as the austenite phase. An increase in the temperature of the SMA induces shape recovery even under significant applied loads, resulting in high actuation energy densities (work output per unit mass) compared to other active materials with direct coupling. Under certain conditions, SMAs can absorb and dissipate mechanical energy through a reversible hysteretic shape change when subjected to cyclic mechanical loading. These unique properties make SMAs highly suitable for sensing, vibration damping, and especially for actuation applications.

[0058] The function of SMA coils is based on the shape memory effect found in nickel-titanium alloys. The concentration of elements in the alloy is crucial, especially when it consists of at least two elements. When nickel and titanium atoms are present in nearly equal proportions, the alloy forms a crystal structure that can transform from one form to another at a temperature dependent on the precise composition. In the austenite phase above the transformation temperature, the material exhibits high strength and resists deformation under load, behaving similarly to stainless steel and thus withstanding higher stress during actuation.

[0059] Generally, SMAs have two temperature-dependent phases with distinct crystal structures and properties: the low-temperature phase, martensite (M), and the high-temperature phase, austenite (A). The austenite phase has a cubic crystal structure, while martensite can exhibit tetragonal, orthorhombic, or monoclinic structures. The transformation between these structures is driven by shear lattice distortion, known as the martensitic transformation. This reversible phase transformation between martensite and austenite underpins the unique behavior of SMAs.

[0060] The Shape Memory Effect (SME) is responsible for the macroscopic shape change and its full recovery during phase transformation. Under no load conditions, when the temperature reaches the martensitic start temperature (Ms), the crystal structure begins to transform to twinned martensite and completes the transformation at the martensitic finish temperature (Mf); this is called the forward transformation. At this point, the material is entirely in the twinned martensitic phase. The reverse transformation occurs during heating; it starts at the austenitic start temperature (As) and completes at the austenitic finish temperature (Af). Under applied stress, the low-temperature twinned martensitic phase converts to detwinned martensite due to reorientation, resulting in a macroscopic shape change, where the deformed configuration is retained when the stress is relieved. Subsequent heating of the SMA above Af transitions the phase from detwinned martensite to austenite (reverse phase transformation), leading to complete shape recovery. When the temperature falls below Mf, twinned martensite forms without any associated shape change during the forward transformation.

[0061] Referring to FIG. 2, where an exemplary circuit diagram of the proposed device is disclosed. During the cooling stage, the current supply to the thermally sensitive coil 114 may be switched off, and the humidifier module 128, 134 may be activated by the control unit. The humidifier module 128, 134 may be configured to disperses coolant, initiating the movement of the subassembly back to its original position. This motion may be facilitated by the presence of biased springs 120. This entire sequence of operations constitutes the complete cycle necessary for obtaining a rotary output from the device 100.

[0062] Referring to FIG. 3, where an exemplary graph representing cooling frequency of the thermally sensitive coil 114 using acetone cooling fluid and free convection cooling is disclosed. The use of acetone cooling significantly reduces the cooling time of the thermally sensitive coil 114, thereby decreasing the overall actuation time. This reduction in cooling time enhances the frequency of movement between the first and second positions of the coil holder 110, enabling more rapid and efficient operation. By quickly lowering the temperature

of the SMA coil, acetone facilitates faster phase transformation from austenite to martensite, allowing the system to reset and prepare for the next actuation cycle more swiftly.

[0063] The acetone cooling fluid is stored in designated cooling tanks 130 and 136, from which it is dispersed through one or more humidifier units 128 and 134. These humidifier units ensure an even and controlled distribution of acetone, optimizing the cooling process. The efficient management and application of the cooling fluid are critical to maintaining the performance and reliability of the actuation system. By leveraging the properties of acetone, the system achieves a high frequency of operation, which is essential for applications requiring rapid and repetitive movements.

[0064] While the foregoing describes various embodiments of the disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof. The scope of the disclosure is determined by the claims that follow. The disclosure is not limited to the described embodiments, versions, or examples, which are included to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art.

ADVANTAGES OF THE PRESENT DISCLOSURE

[0065] The present disclosure provides an improved device that converts reciprocating motion into rotational motion in higher torque.

[0066] The present disclosure produces higher actuation frequency by reducing actuation timing of the SMA or thermal sensitive coil.

[0067] The present disclosure adapts for various industrial applications, including power generation, robotics, and automated machinery.

[0068] The present disclosure provides high power to weight ratio.

[0069] The present disclosure minimizes noise for being adapted to sensitive environments.

We Claim:

1. A device (100) for converting reciprocating motion into rotational motion, the device (100) comprising;

a housing (102) having a base platform (104);

a rotary member (106) pivotally coupled to the centre of the base platform (104), the rotary member (106) having at least one free end (106-1), wherein the rotary member (106) is configured to move between a first position and a second position;

a plurality of shape memory alloy (SMA) coils (114) coupled between the at least one coil holder (110) and the rotary member (106), wherein the plurality of shape memory alloy (SMA) coils (114) are configured to contract in response to an increase in temperature supplied by a heating unit and to return towards an extended state upon cooling;

a cooling unit (116) positioned on the base platform (104);

characterized in that

at least one connecting link (108) coupled to at least one coil holder (110) through an attachment means (112);

wherein the at least one coil holder (110) comprises a first coil holder (110-1) and a second coil holder (110-2), each coil holder (110) rotatably coupled to the second end (108-lB, 108-2B) of each of the at least one connecting link (108) for converting reciprocating movement of the at least one coil holder (110) into rotary motion using the at least one connecting link (108);

wherein the cooling unit (116) is configured to spray cooling fluid onto the heated shape memory alloy (SMA) coils (114) through the one or more humidifier units (128,134) to reduce cooling time of the shape memory alloy (SMA) coils (114),

wherein the cooling unit (116) comprises a first cooling unit (116-1) and a second cooling unit (116-2) fluidically coupled to a cooling fluid tank (130,136) through one or more conduits (132),

wherein the at least one coil holder (110) is configured on one or more guide bars (118) that allow slidable movement between the first position and the second position in response to the contraction and expansion of the plurality of shape memory alloy (SMA) coils (114), the movement of the at least one coil holder (110) being transmitted to the rotary member (106)

through the at least one connecting link (108) such that the plurality of shape memory alloy (SMA) coils (114) and the cooling unit (116) facilitate rotation of the rotary member (106).

2. The device (100) as claimed in claim 1, wherein the rotary member (106) comprises at least two free ends (106-1, 106-2) and the at least one connecting link (108) comprises at least a pair of connecting links (108-1, 108-2), wherein each of the free ends (106-1, 106-2) of the rotary member (106) is coupled to the first end (108-1A, 108-2A) of each of the at least one connecting link (108).

3. The device (100) as claimed in claim 1, wherein the device (100) comprises a first set of bias springs (120) positioned parallel to the plurality of shape memory alloy (SMA) coils (114), the first set of bias springs (120) having a first end (120-1) and a second end (120-2), the first end (120-1) of the first set of bias springs (120) coupled to the first coil holder (110-1) and the second end (120-2) of the first set of bias springs (120) coupled to the second coil holder (110-2) through one or more strings (122), wherein the first set of bias springs (120) is configured to restrain the first and second coil holder (110-1, 110-2) in the second position due to temperature changes in the thermal sensitive coil (114).

4. The device (100) as claimed in claim 3, wherein the one or more string (122) is supported by one or more pulleys (124) positioned on the base platform (104), the one or more pulleys (124) configured to accommodate a portion of the string (122) to facilitate movement of the coil holder (110).

5. The device (100) as claimed in claim 1, wherein the device (100) comprises one or more bias springs (126) configured on the one or more guide rods (118), the bias springs (126) having a first end (126-1) and a second end (126-2), the first end (126-1) of the bias springs (126) coupled to the first coil holder (110-1) and the second end (126-2) of the bias springs (126) coupled to the rotary member (106), wherein the one or more bias springs (126) are configured to restrain the first and second coil holder (110-1, 110-2) in the second position due to temperature changes in the shape memory alloy (SMA) coil (114).

6. The device (100) as claimed in claim 1, wherein the heating unit is located in the housing (102) and configured to supply power to the shape memory alloy (SMA) coil (114) to contract and facilitate movement of the first and second coil holders (110-1, 110-2) towards the first position.

7. The device (100) as claimed in claim 1, wherein the at least one connecting link (108) comprises a first end (108-1) and a second end (108-2), wherein the first end (108-1) of the at

least one connecting link (108) coupled to a free end (106-1) of the rotary member (106) and the second end (108-2) of the at least one connecting link (108) coupled to at least one coil holder (110) through an attachment means (112).

8. The device (100) as claimed in claim 1, wherein the rotary member (106) comprises an output shaft (140) configured to rotate along with the rotary member (106), the output shaft (140) coupled to a load or device for transferring rotational motion.

FOR Indian Institute of Technology Kanpur



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Agent for the Applicant

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Application Number - 202411043335 Granted Specification - Page 17

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TARUN KHURANA
Date: 30-04-2026 17:26:53

ABSTRACT

A SHAPE MEMORY ALLOY-BASED DEVICE WITH ENHANCED ROTATIONAL MOTION

A shape memory alloy-based device with enhanced rotational motion (100) for converting reciprocating motion into rotational motion that includes a housing (102) with a base platform (104) and is a rotary member pivotally coupled on the base platform (104) and configured to move between a first and a second position; at least one connecting link (108) coupled to the rotary member (106) and a coil holder (110) through an attachment means (112); a thermally sensitive coil (114) positioned between the coil holder (110) and the rotary member (106), contracting and expanding with temperature changes induced by cooling (116) and heating units. The coil holder (110) is configured on guide bars (118), enabling slidable movement between the two positions which is transferred to the rotary member (106) through the connecting link (108), converting the linear motion of the coils (114) into rotational motion with increased torque and increased frequency.

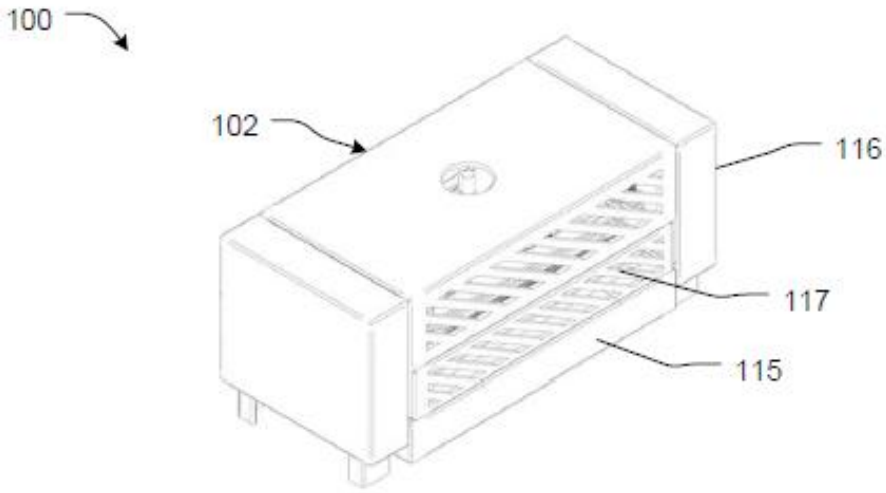


FIG. 1A

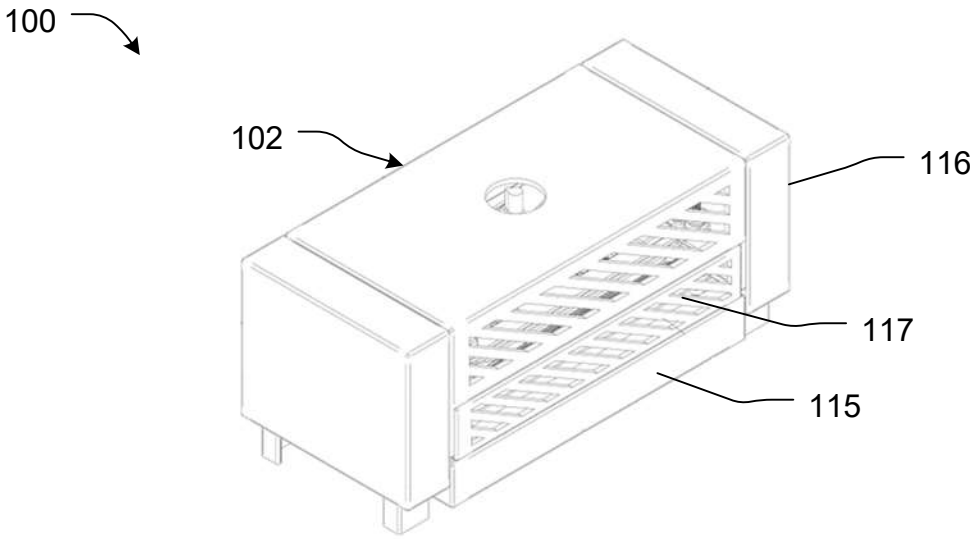


FIG. 1A

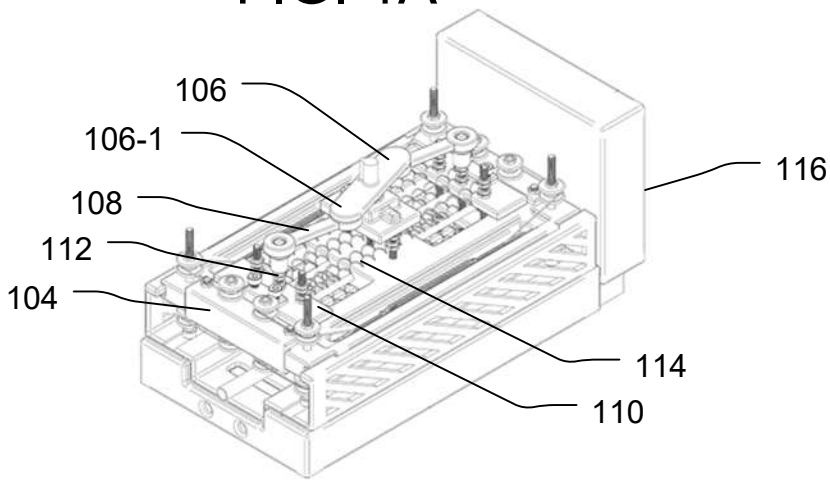


FIG. 1B

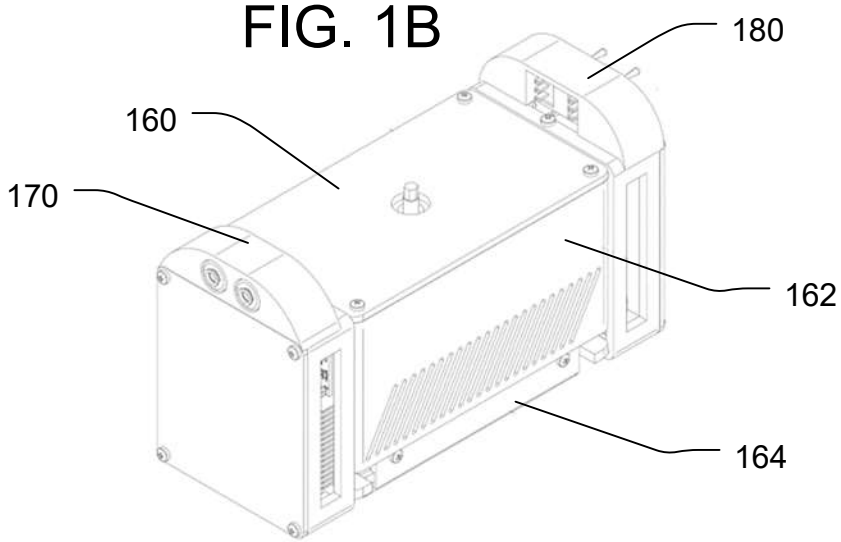


FIG. 1C

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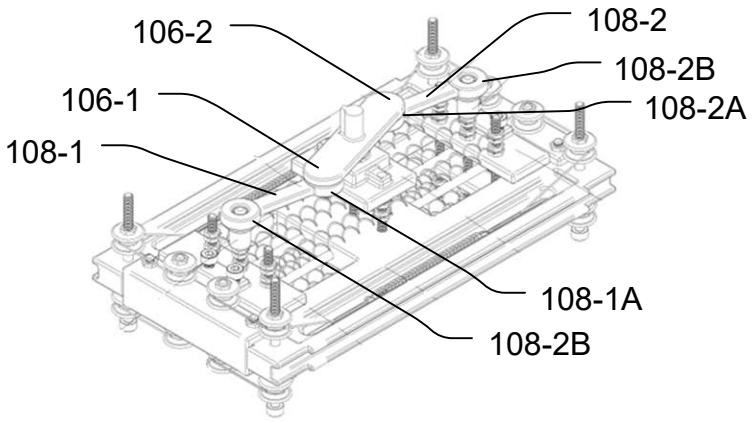


FIG. 1D

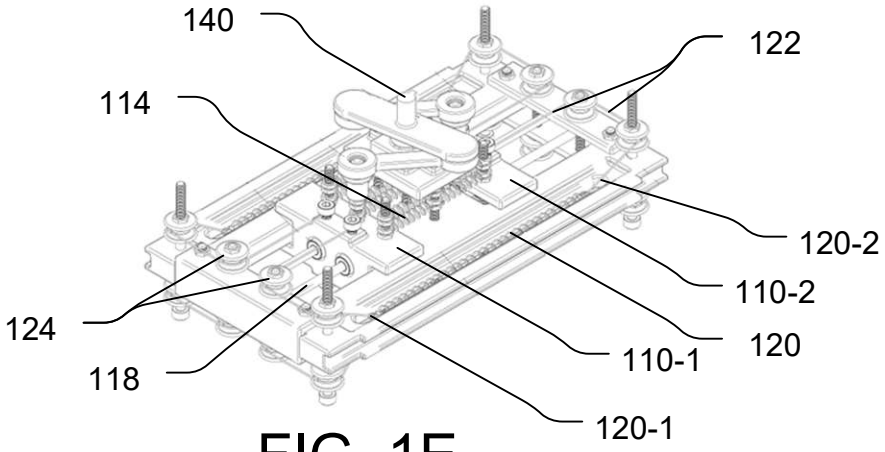


FIG. 1E

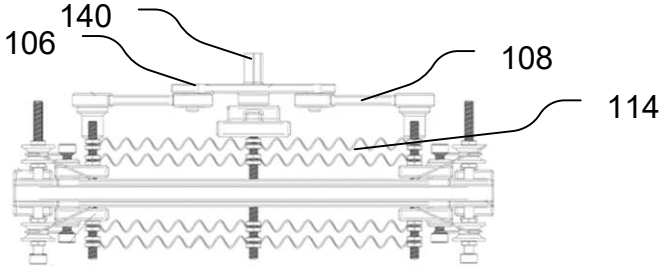


FIG. 1F

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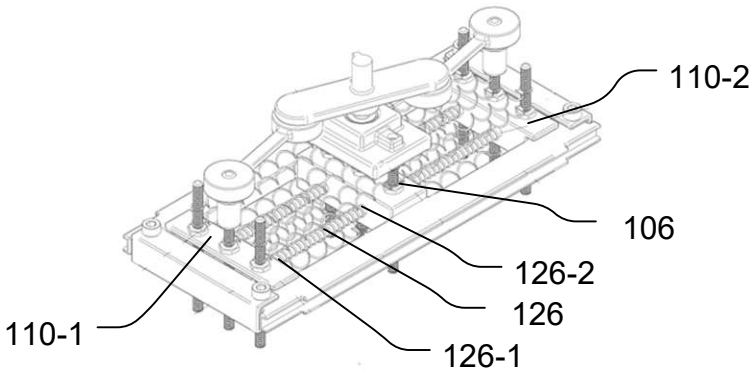


FIG. 1G

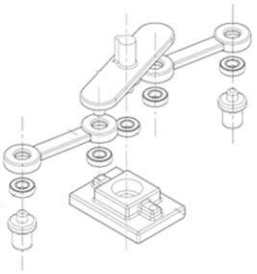


FIG. 1H

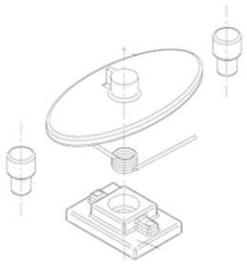


FIG. 1I

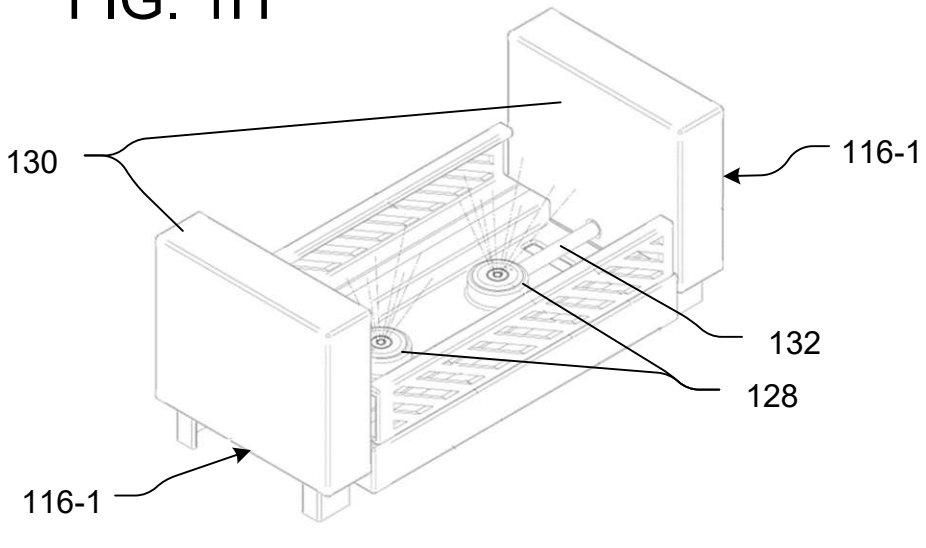


FIG. 1J

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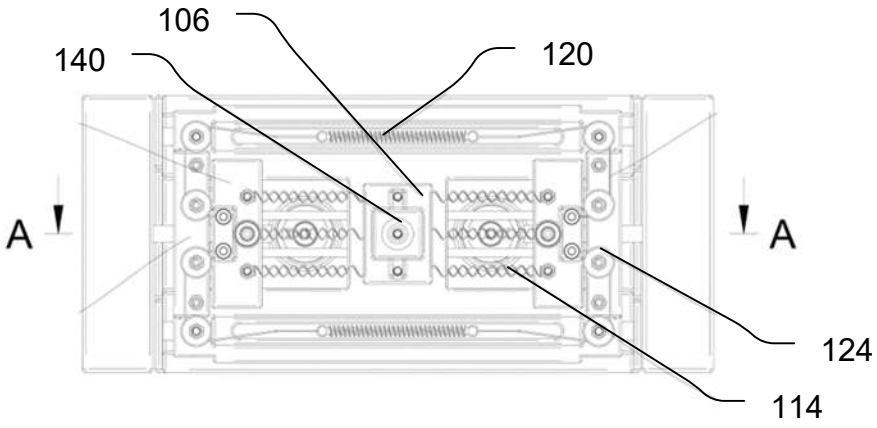


FIG. 1K

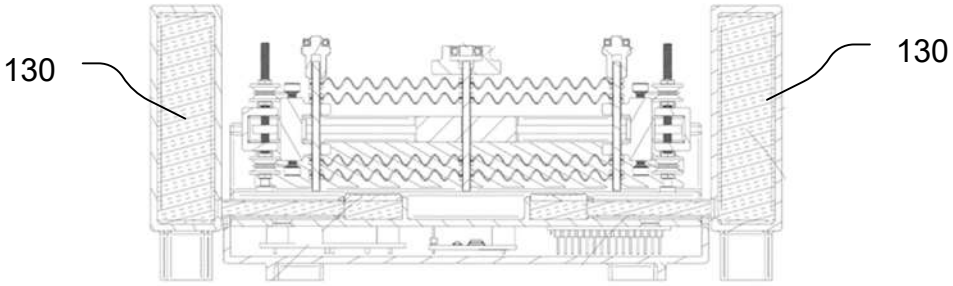


FIG. 1L

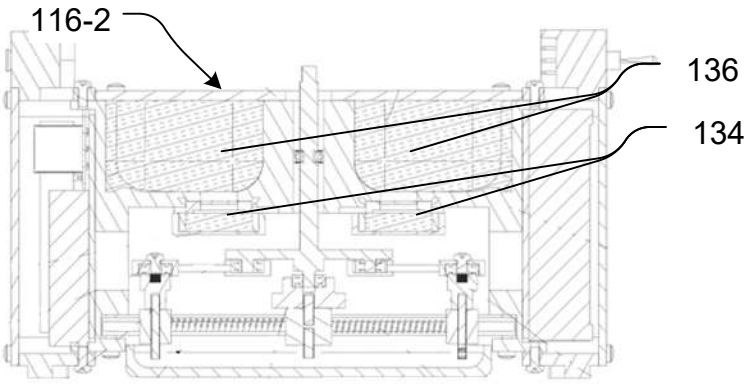


FIG. 1M

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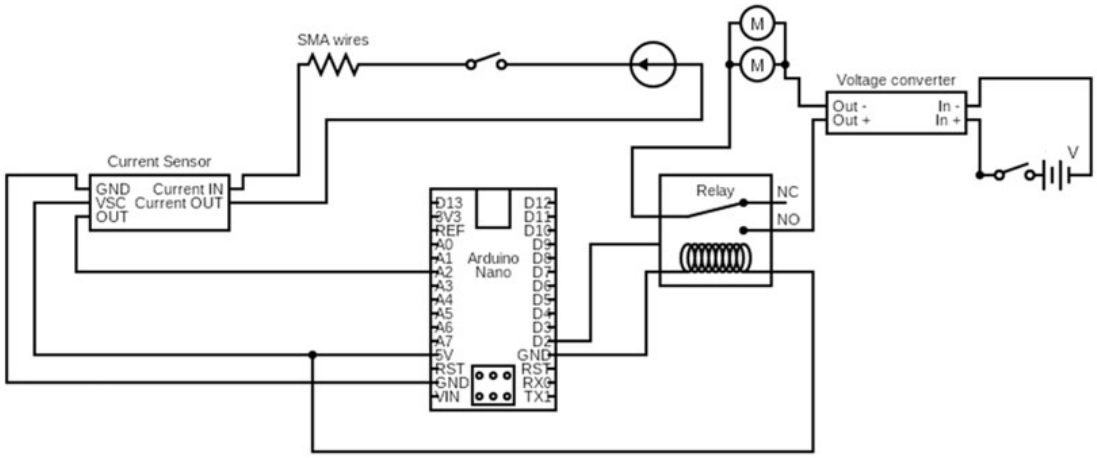


FIG. 2

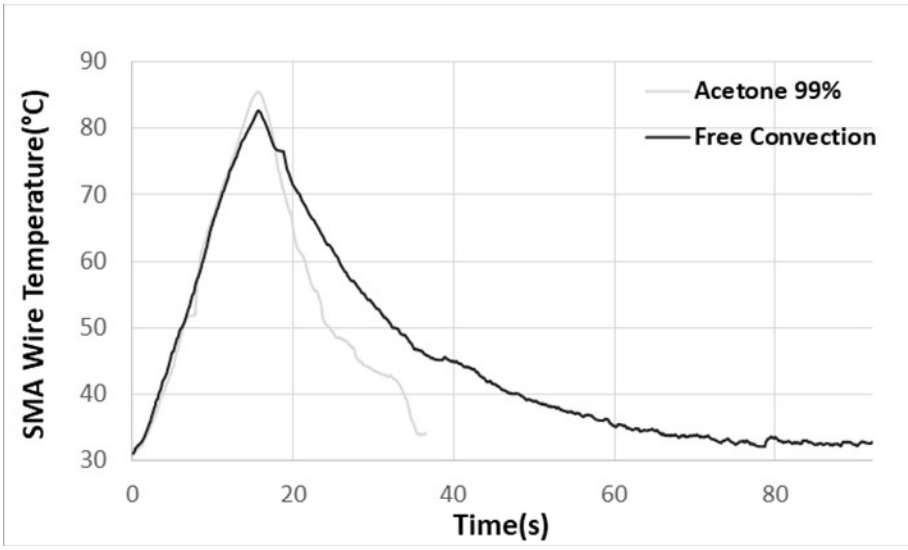


FIG. 3

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Regd. Patent Agent [IN/PA-1325]
Dated: 04th June, 2024